

Chapter 6 — Best practices in EDA

Exploration is also communication and reproducibility

Learning objectives

- Select charts by analysis type, avoid misleading visual design, document EDA steps
- Tailor presentations to technical versus non-technical audiences

Linking detection to repair

- When EDA surfaces defects, log what you found and apply Chapter 5 repairs
- Short notes tying plots to cleaning actions support reproducibility and later model audits

Choosing the right chart

- Histograms and box plots for univariate numerics
- Simplicity beats decorative complexity

Avoiding misleading visuals

- Do not truncate axes to exaggerate differences
- Anscombe's quartet reminds us identical summaries can hide different shapes

Reproducibility

- Document steps in notebooks or scripts; explain why transforms or charts were chosen
- Use version control for analysis artifacts
- Markdown cells in Jupyter should state the question each plot answers

Collaboration and storytelling

- Tailor depth to the audience
- Each chart should convey one message

Takeaways

- Good EDA is reproducible and honest
- Chart choice and annotation matter as much as code
- Link exploration logs to cleaning and modeling downstream

Next

- Complete the quiz for this part
- Continue to common pitfalls in EDA